

Logic - Well Founded Induction

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1 Induction on $\mathbb{N}$

1.1 What is $\mathbb{N}$?

Definition 1.1 (Peano-Like). *A structure $(N, S_N, 0_N)$ is called **Peano-Like** when*

(PL0) N is a set

(PL1) $0_N \in N$

(PL2) $S_N : N \rightarrow N$

(PL3) S_N is injective

(PL4) $\forall n \in N. S_N(n) \neq 0_N$

(PL5) $\forall n \in N. n = 0_N \vee \exists k \in N. (S_N(k) = n)$

We write $n + 1 := S_N(n)$.

Axiom 1.2. *There exists a Peano-Like structure. We fix one and call it $(\mathbb{N}, S, 0)$.*

Nothing in (PL0) to (PL5) says that induction holds. The conatural numbers $\mathbb{N} \cup \{\infty\}$ with $S(\infty) = \infty$ satisfy all six conditions and violate both propositions below. Everything in this section is therefore proved for an arbitrary Peano-Like structure, and $\mathbb{N}$ can be read as any one of them.

Remark (on the choice of axioms). The proof of Theorem 1.5 below uses (PL5) in the $\rightarrow$ direction, where we split n into the case $n = 0$ and the case $n = S(k)$. There were other ways to set things up.

Option 1: drop (PL5). Then the equivalence is false, not just harder to prove. Let X be two disjoint copies of the natural numbers, with S the usual successor on each copy and 0 the bottom of the first copy. Then (PL0) to (PL4) hold, but the bottom c of the second copy is not 0 and has no predecessor. Proposition 1.3 fails: take B to be the first copy, which contains 0 and is closed under S but is not all of X . Proposition 1.4

holds: any B satisfying its hypothesis contains 0 and c , since both have $s[\cdot] = \emptyset$, and then contains every successor of an element of B , hence both copies. So Proposition 1.4 holds and Proposition 1.3 fails in this structure. The same B shows that the pointwise equivalence used in the proof below fails, since $s[c] = \emptyset \subseteq B$ while $c \notin B$.

Option 2: use the usual Peano axioms with induction as an axiom. Take (PL0) to (PL4) together with Proposition 1.3. Then (PL5) becomes a lemma, by applying induction to $\{0\} \cup S[\mathbb{N}]$, which contains 0 and is closed under S . This setup makes the equivalence less informative, since the induction principle is assumed instead of derived from the structure. (PL3) and (PL4) are still needed, but in the $\leftarrow$ direction of the proof below.

My choice. I keep (PL5) as an axiom of the structure. This isolates what the equivalence really depends on, and it allows the direct proof by reducing the goal for each fixed B .

1.2 The induction statement on $\mathbb{N}$

Proposition 1.3. *For every $B \subseteq \mathbb{N}$*

$$[0 \in B \wedge \forall n \in \mathbb{N}. (n \in B \rightarrow n + 1 \in B)] \rightarrow B = \mathbb{N}$$

Proposition 1.4. *For every $B \subseteq \mathbb{N}$*

$$[\forall n \in \mathbb{N}. (s[n] \subseteq B \rightarrow n \in B)] \rightarrow B = \mathbb{N}$$

Theorem 1.5.

$$\text{Prop 1.3} \longleftrightarrow \text{Prop 1.4}$$

Lemma 1.6 (Reduction). *Let X be a set and $\varphi_1, \varphi_2, \psi$ be predicates on X with $\forall x \in X. (\varphi_1(x) \longleftrightarrow \varphi_2(x))$. Then*

$$\forall x \in X. (\varphi_1(x) \rightarrow \psi(x)) \longleftrightarrow \forall x \in X. (\varphi_2(x) \rightarrow \psi(x)).$$

Proof. For $\rightarrow$, assume the left side and take x with $\varphi_2(x)$. Then $\varphi_1(x)$ by the assumed equivalence, so $\psi(x)$. The $\leftarrow$ direction is the same with φ_1 and φ_2 exchanged. $\square$

Proof of Theorem 1.5. We apply Lemma 1.6 with $X = \mathcal{P}(\mathbb{N})$, $x = B$, $\psi(B) := (B = \mathbb{N})$,

$$\varphi_1(B) := 0 \in B \wedge \forall n \in \mathbb{N}. (n \in B \rightarrow n + 1 \in B),$$

$$\varphi_2(B) := \forall n \in \mathbb{N}. (s[n] \subseteq B \rightarrow n \in B).$$

Since ψ is the same on both sides, it is enough to show $\varphi_1(B) \longleftrightarrow \varphi_2(B)$ for every $B \subseteq \mathbb{N}$.

Take arbitrary $B \subseteq \mathbb{N}$.

Firstly, $\varphi_1(B) \rightarrow \varphi_2(B)$. Assume

$$0 \in B \wedge \forall n \in \mathbb{N}. (n \in B \rightarrow n + 1 \in B). \quad (1)$$

W.T.S¹

$$\forall n \in \mathbb{N}. ({}_S[n] \subseteq B \rightarrow n \in B).$$

Take arbitrary $n \in \mathbb{N}$ with ${}_S[n] \subseteq B$. We must show $n \in B$. By (PL5), either $n = 0$ or $n = S(k)$ for some k .

Case $n = 0$. By the left side of (1), $n = 0 \in B$.

Case $n = S(k)$ for some $k \in \mathbb{N}$. Then $k \in {}_S[n]$, and ${}_S[n] \subseteq B$ gives $k \in B$. By the right side of (1), $k + 1 \in B$. But $k + 1 = S(k) = n$, so $n \in B$.

Secondly, $\varphi_2(B) \rightarrow \varphi_1(B)$. Assume

$$\forall n \in \mathbb{N}. ({}_S[n] \subseteq B \rightarrow n \in B). \quad (2)$$

W.T.S

$$0 \in B \wedge \forall n \in \mathbb{N}. (n \in B \rightarrow n + 1 \in B).$$

$0 \in B$?

By (PL4), no k has $S(k) = 0$, so ${}_S[0] = \emptyset \subseteq B$. By (2), $0 \in B$.

$\forall n \in \mathbb{N}. (n \in B \rightarrow n + 1 \in B)$?

Take arbitrary $n \in \mathbb{N}$ with $n \in B$. We want $n + 1 \in B$, and by (2) applied to $n + 1$ it suffices to show ${}_S[n + 1] \subseteq B$. Take any $k \in {}_S[n + 1]$. Then $S(k) = n + 1 = S(n)$, so $k = n$ by (PL3). But $n \in B$, so $k \in B$. Hence ${}_S[n + 1] \subseteq B$.

So $\varphi_1(B) \longleftrightarrow \varphi_2(B)$ for every $B \subseteq \mathbb{N}$, and Lemma 1.6 gives the theorem. $\square$

2 Well-Founded induction

2.1 On infinite descending sequences

Definition 2.1 (Infinite descending ρ -sequence). *a is called an **infinite descending** ρ -sequence if it is an infinite sequence such that*

$$\forall n \in \mathbb{N}. (a_{n+1} \rho a_n).$$

Definition 2.2 (Range). *Let $a : \mathbb{N} \rightarrow X$ a set $A \subseteq X$ is called the range of a such that*

$$A = \{x \in X \mid \exists n \in \mathbb{N}. (a_n = x)\},$$

and could be wrote $A = \text{Range } a$.

¹W.T.S := Want to Show

Theorem 2.3. Let X be a set and $a : \mathbb{N} \rightarrow X$ then

$$\text{Range } a \neq \emptyset.$$

Proof. It is enough to prove $\exists x \in X. (x \in \text{Range } a)$.

Take $x = a_0$, then $\exists n \in \mathbb{N}. (a_n = x)$ which is true by $n = 0$.

So, $x \in \text{Range } a$. □

Theorem 2.4. Let a be an infinite descending ρ -sequence then

$$\forall x \in \text{Range } a. (\rho[x] \cap \text{Range } a \neq \emptyset)$$

Proof. Take arbitrary $x \in \text{Range } a$, then $\exists n \in \mathbb{N}. (a_n = x)$, so take such one and call it n' , i.e, $x = a_{n'}$

Since $\forall n \in \mathbb{N}. (a_{n+1} \rho a_n)$, so apply it for our n' , hence we have

$$a_{n'+1} \rho x.$$

Let $y := a_{n'+1}$. Is $y \in \text{Range } a$? So we need to show $\exists n \in \mathbb{N}. (a_n = y)$, which is true by definition with taking $n = n' + 1$.

Is $y \in \rho[x]$? So it is enough to show $y \rho x$, which also true by definition by virtue of

$$(y = a_{n'+1}) \rho (a_{n'} = x).$$

Thus, $y \in \rho[x] \cap \text{Range } a$, so it is not empty. □

Definition 2.5. (Well Founded) we call a relation ρ **well founded** if for every $B \neq \emptyset$

$$\exists x \in B. (\rho[x] \cap B = \emptyset)$$

Definition 2.6 (Principle of ρ , A -induction). For a set A , the principle of ρ , A -induction is: for every $B \subseteq A$

$$[\forall x \in A. (\rho[x] \cap A \subseteq B \rightarrow x \in B)] \rightarrow B = A.$$

Lemma 2.7. If $B \subseteq A$ and $B \neq A$, then $A \setminus B \neq \emptyset$.

Proof. Suppose $A \setminus B = \emptyset$. Take $x \in A$. If $x \notin B$ then $x \in A \setminus B$, which is impossible, so $x \in B$. Hence $A \subseteq B$, and with $B \subseteq A$ we get $B = A$, a contradiction. □

Lemma 2.8. If $C \cap (A \setminus B) = \emptyset$ then $C \cap A \subseteq B$.

Proof. Take $x \in C \cap A$. If $x \notin B$ then $x \in C \cap (A \setminus B)$, contradicting the hypothesis. So $x \in B$. □

Theorem 2.9. The following are equivalent.

(i) ρ is well founded.

(ii) For every A , the principle of ρ , A -induction holds.

(iii) *There is no infinite descending ρ -sequence.*

Proof. (i) $\rightarrow$ (ii). Assume (i). Let A be a set and $B \subseteq A$ with

$$\forall x \in A. (\rho[x] \cap A \subseteq B \rightarrow x \in B). \quad (3)$$

Suppose $B \neq A$. By Lemma 2.7, $A \setminus B \neq \emptyset$, so (i) gives $y \in A \setminus B$ with $\rho[y] \cap (A \setminus B) = \emptyset$. By Lemma 2.8, $\rho[y] \cap A \subseteq B$. Then (3) applied to $y \in A$ gives $y \in B$, contradicting $y \notin B$. So $B = A$.

(ii) $\rightarrow$ (iii). Assume (ii) and suppose a is an infinite descending ρ -sequence. Let $A = \text{Range } a$ and $B = \emptyset$. Then $B \subseteq A$. We check (3). Take $x \in A$ with $\rho[x] \cap A \subseteq \emptyset$. Then $\rho[x] \cap A = \emptyset$, which contradicts the theorem 2.4 on $\text{Range } a$. So the implication in (3) holds vacuously. By (ii), $\emptyset = \text{Range } a$, but $\text{Range } a \neq \emptyset$, a contradiction.

(iii) $\rightarrow$ (i). We prove the contrapositive. Assume ρ is not well founded. Then there is $B \neq \emptyset$ such that

$$\forall x \in B. (\rho[x] \cap B \neq \emptyset).$$

So every $x \in B$ has some $y \in B$ with $y \rho x$. By the axiom of choice there is $f : B \rightarrow B$ with $f(x) \rho x$ for all $x \in B$. Pick $x_0 \in B$ (since B is not empty) and define $a : \mathbb{N} \rightarrow B$ by recursion:

$$a_0 = x_0, \quad a_{n+1} = f(a_n).$$

Then $a_{n+1} = f(a_n) \rho a_n$ for every n , so a is an infinite descending ρ -sequence, and (iii) fails. $\square$

Definition 2.10. We say ρ is *well founded on* A if for every nonempty $B \subseteq A$

$$\exists x \in B. (\rho[x] \cap B = \emptyset).$$

Proposition 2.11. Let $R_\rho = \{y \mid \exists x. (x \rho y)\}$. ρ is well founded iff ρ is well founded on R_ρ .

Proof. ($\Rightarrow$) A nonempty $B \subseteq R_\rho$ is in particular a nonempty set, so well-foundedness of ρ applies to it directly.

($\Leftarrow$) Let $B \neq \emptyset$. We split into two cases.

Case 1: some $x \in B$ has $x \notin R_\rho$. Then $\rho[x] = \emptyset$: if $y \in \rho[x]$ then $y \rho x$, so $x \in R_\rho$. Hence $\rho[x] \cap B = \emptyset$.

Case 2: every $x \in B$ lies in R_ρ . Then $B \subseteq R_\rho$ is nonempty, and the hypothesis gives the required $x \in B$. $\square$

Proposition 2.12. For every A the principle of ρ , A -induction holds iff the principle of ρ , R_ρ -induction holds.

Proof. ($\Rightarrow$) Take $A = R_\rho$.

($\Leftarrow$) By Theorem 2.9 and Proposition 2.11 it is enough to show that ρ is well founded on R_ρ . Let $B \subseteq R_\rho$ be nonempty and suppose no $x \in B$ has $\rho[x] \cap B = \emptyset$. Put $C = R_\rho \setminus B$. We apply ρ , R_ρ -induction to C .

Take $x \in R_\rho$ with ${}_\rho[x] \cap R_\rho \subseteq C$. We show $x \in C$, which needs $x \notin B$. Suppose $x \in B$. By assumption ${}_\rho[x] \cap B \neq \emptyset$, so there is $y \in {}_\rho[x] \cap B$. Since $B \subseteq R_\rho$, we get $y \in {}_\rho[x] \cap R_\rho \subseteq C$, so $y \notin B$, a contradiction.

By induction $C = R_\rho$. Take $x \in B$. Then $x \in R_\rho = C$, so $x \notin B$, a contradiction. Hence ρ is well founded on R_ρ . $\square$